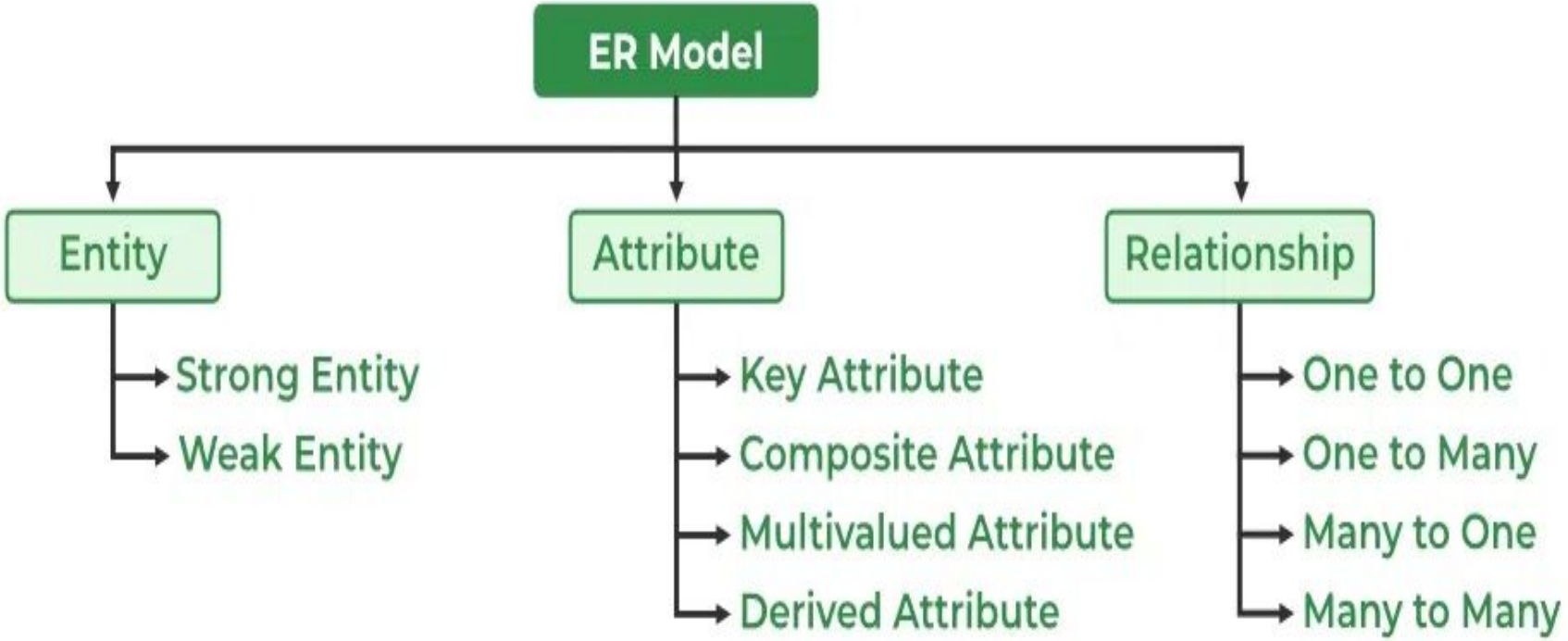
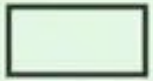
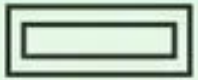
Entity Relationship Model (ER- Model)

Unit 2

Introduction of ER Model

The Entity-Relationship Model (ER Model) is a conceptual model for designing a databases. This model represents the logical structure of a database, including entities, their attributes and relationships between them.


ER Model
Entity
Attribute
Relationship
Strong Entity
Weak Entity
Key Attribute
Composite Attribute
Multivalued Attribute
Derived Attribute
One to One
One to Many
Many to One
Many to Many

Figures	Symbols	Represents
Rectangle		Entities in ER Model
Ellipse		Attributes in ER Model
Diamond		Relationships among Entities
Line		Attributes to Entities and Entity Sets with Other Relationship Types
Double Ellipse		Multi-Valued Attributes
Double Rectangle		Weak Entity

Entity

An Entity represents a real-world object, concept or thing about which data is stored in a database. It act as a building block of a database. Tables in relational database represent these entities.

Example of entities:

- **Real-World Objects:** Person, Car, Employee etc.
- **Concepts:** Course, Event, Reservation etc.
- **Things:** Product, Document, Device etc.

Types of Entity

1. Strong Entity

A Strong Entity is a type of entity that has a key Attribute that can uniquely identify each instance of the entity. A Strong Entity does not depend on any other Entity in the Schema for its identification. It has a primary key that ensures its uniqueness and is represented by a rectangle in an ER diagram.

2. Weak Entity

A Weak Entity cannot be uniquely identified by its own attributes alone. It depends on a strong entity to be identified.. A weak entity are represented by a double rectangle.

Attributes in ER Model

Attributes are the properties that define the entity type. For example, for a Student entity Roll_No, Name, DOB, Age, Address, and Mobile_No are the attributes that define entity type Student. In ER diagram, the attribute is represented by an oval.



Types of Attributes

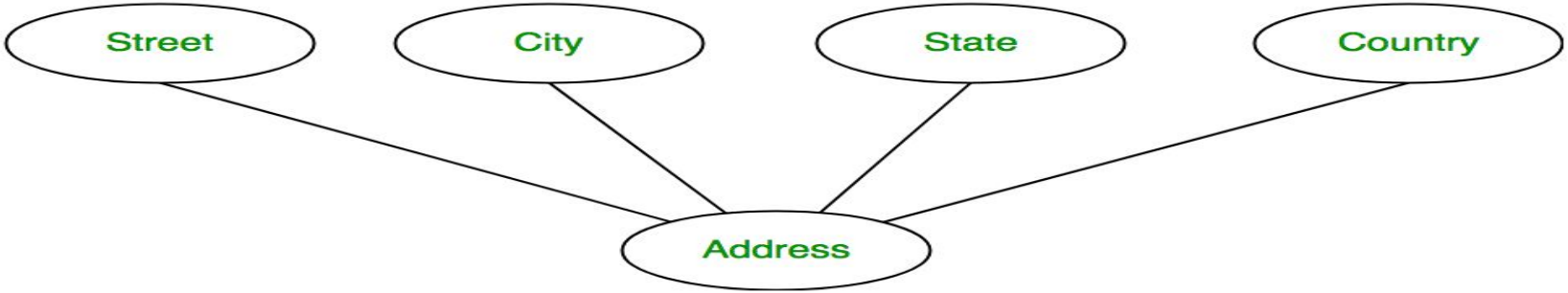
1. Key Attribute

The attribute which uniquely identifies each entity in the entity set is called the key attribute. For example, Roll_No will be unique for each student. In ER diagram, the key attribute is represented by an oval with an underline.



2. Composite Attribute

An attribute composed of many other attributes is called a composite attribute. For example, the Address attribute of the student Entity type consists of Street, City, State, and Country. In ER diagram, the composite attribute is represented by an oval comprising of ovals.



3. Multivalued Attribute

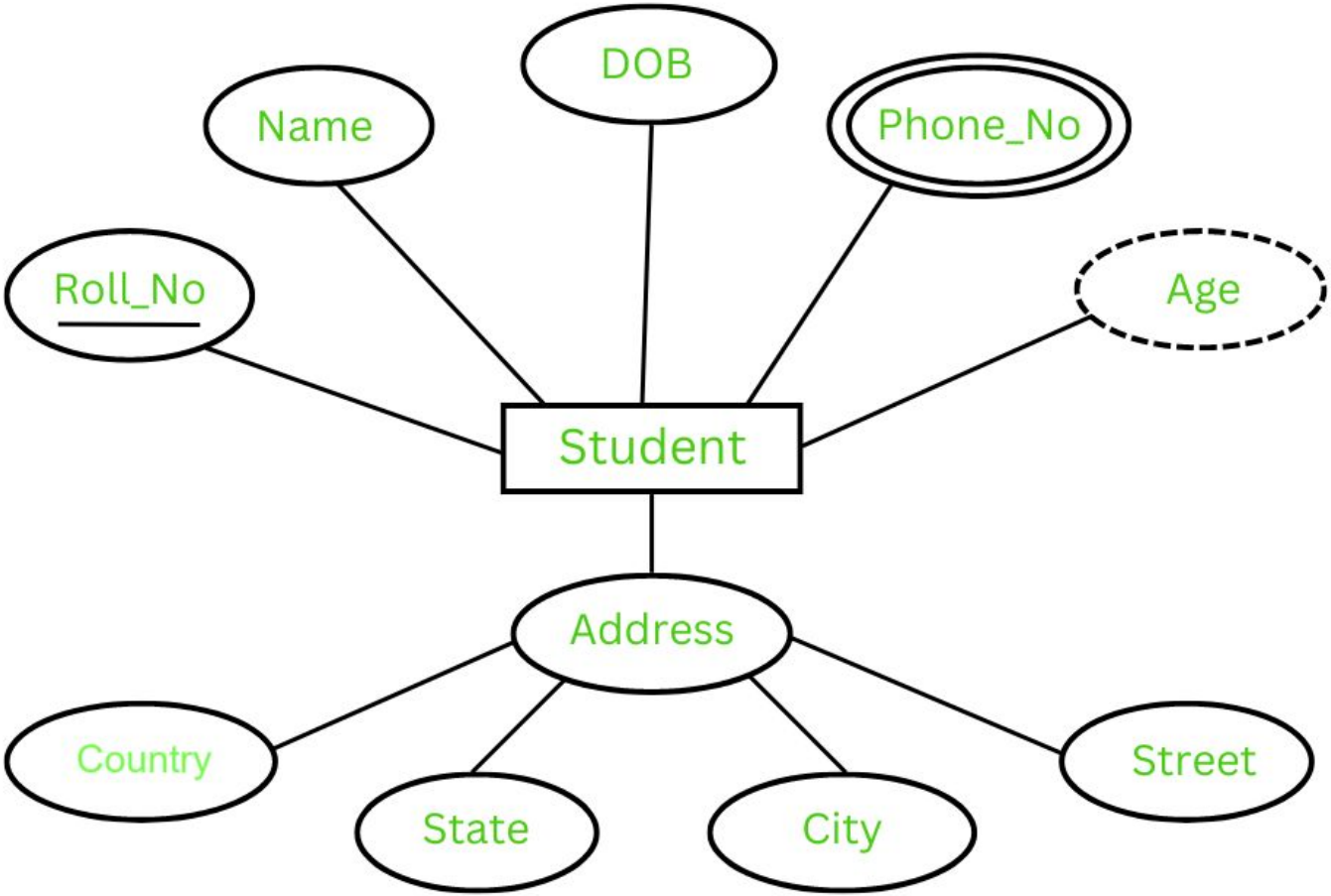
An attribute consisting of more than one value for a given entity. For example, Phone_No (can be more than one for a given student). In ER diagram, a multivalued attribute is represented by a double oval.



4. Derived Attribute

An attribute that can be derived from other attributes of the entity type is known as a derived attribute. e.g.; Age (can be derived from DOB). In ER diagram, the derived attribute is represented by a dashed oval.




DOB
Name
Phone_No
Roll_No
Age
Student
Address
Country
State
City
Street

Relationship in DBMS

A relationship in DBMS is association between two or more entities. Connection between different entities are represented by a diamond shapes in ER (entity relationship) diagrams. Relationships enables the separation and organization of data across multiple tables which facilitate the data management and data retrieval.

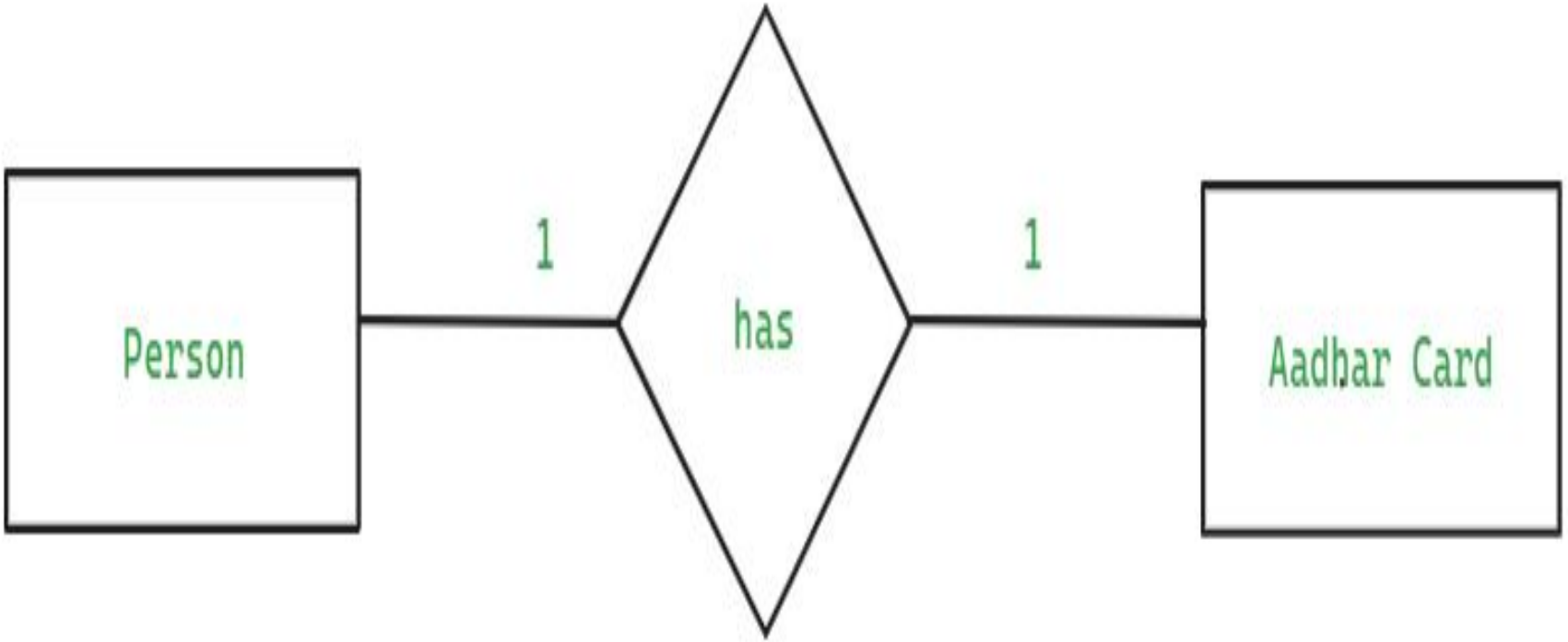
Types of Relationships in DBMS

One-to-One Relationship

In one-to-one relationship a single record in one table is related with a single record in other table and vice versa. This type of relationships are relatively rare and commonly used for security or organizational reasons.

Example:

Consider two entities "Person" and "Aadhar card". Each person can have only one Aadhar card and each Aadhar card is assigned to only one person.

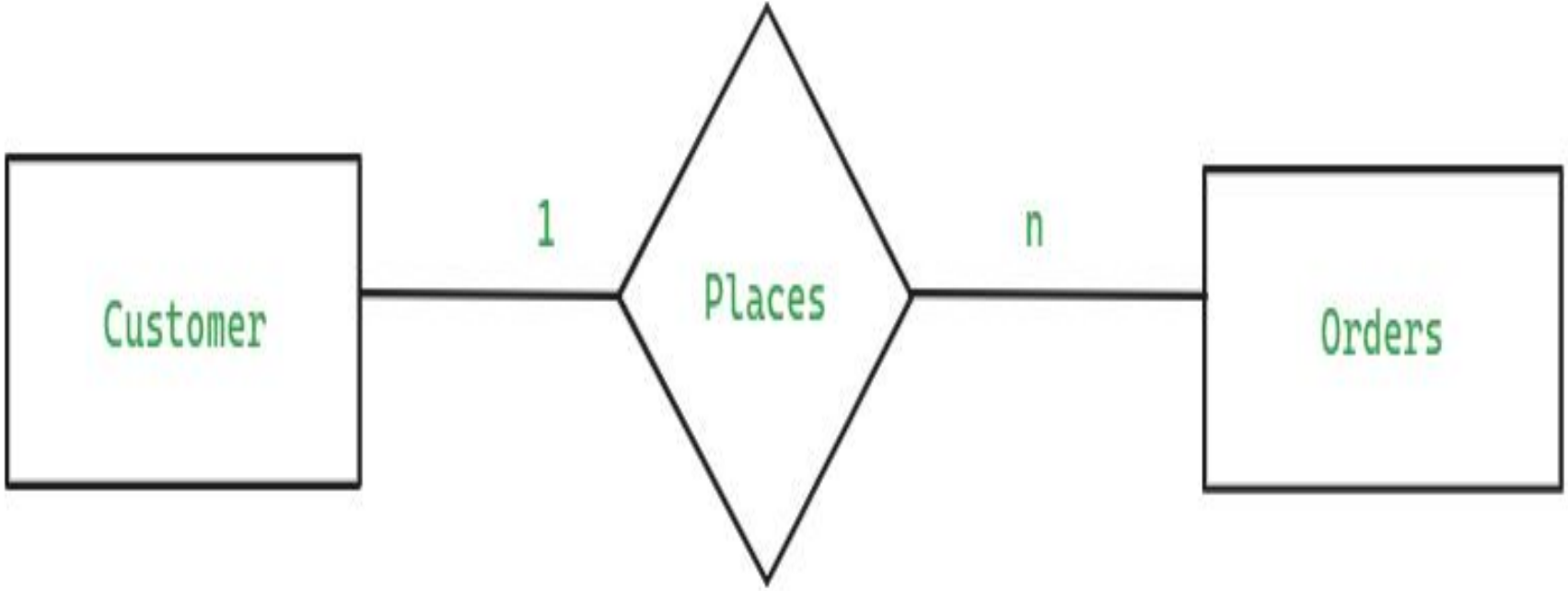

Person
1
has
1
Aadhar Card

One-to-Many or Many-to-One Relationship

In one-to-many or many-to-one relationship, a single record in one table can be associated with multiple records in another table and this is the most common type of relationship in DBMS.

Example:

Consider two entities "customer" and "order". Each customer can place multiple orders but each order is placed by only one customer.

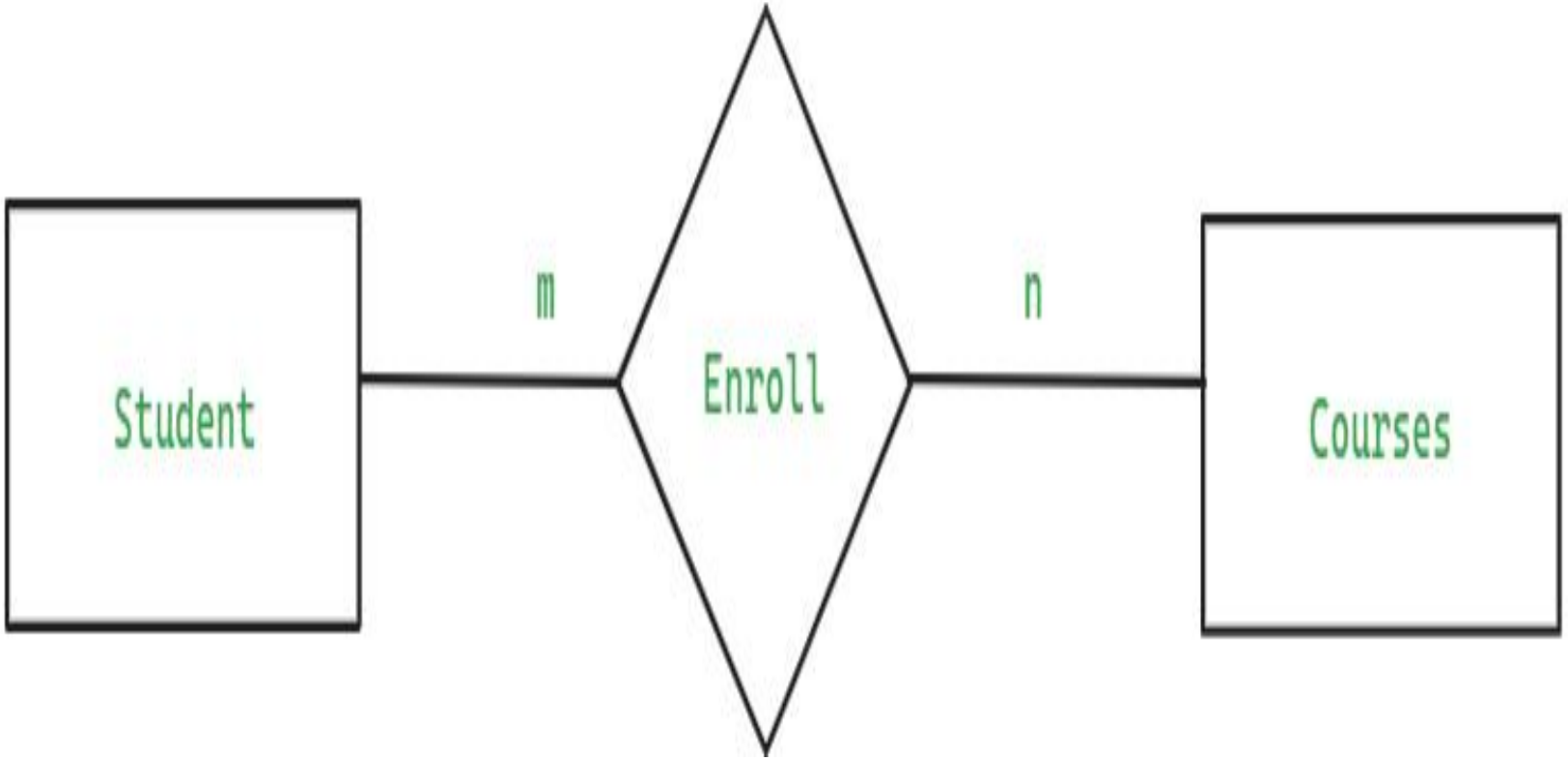

Customer
1
Places
n
Orders

Many-to-Many Relationship

A many-to-many relationship is relationship in which one multiple records in one table are associated with multiple records in another table.

Example:

Consider two entities "Student" and "Course" where each student can enroll in multiple courses and each course can have multiple students enrolled in it.


Student
m
Enroll
n
Courses

Self-Referencing Relationships

A self-referencing relationship is also known as recursive relationship and it is useful is cases when a table has relationship with itself. It is used for representing hierarchical data.

Example:

An "Employee" entity where each employee can have manager who is also an employee.

Mapping Cardinality

It is expressed as the number of entities to which another entity can be associated via a relationship set.

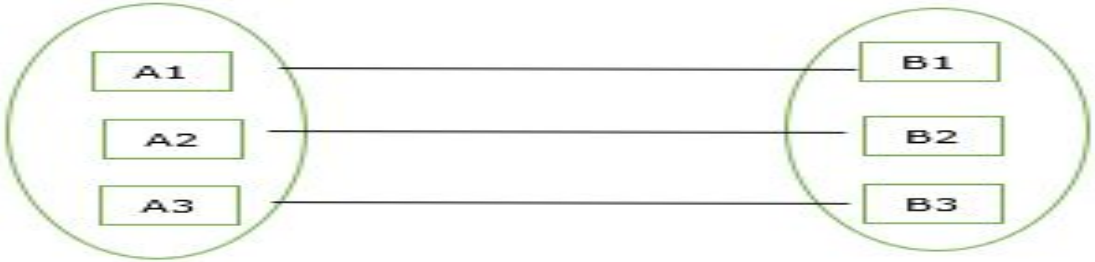
For binary relationship set there are entity set A and B then the mapping cardinality can be one of the following −

- One-to-one
- One-to-many
- Many-to-one
- Many-to-many

One-to-one relationship

One entity of A is associated with one entity of B.

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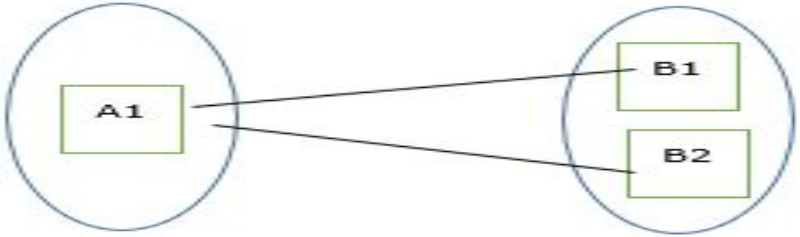
Example

Given below is an example of the one-to-one relationship in the mapping cardinality. Here, one department has one head of the department (HOD).



One-to-many relationship

An entity set A is associated with any number of entities in B with a possibility of zero and entity in B is associated with at most one entity in A



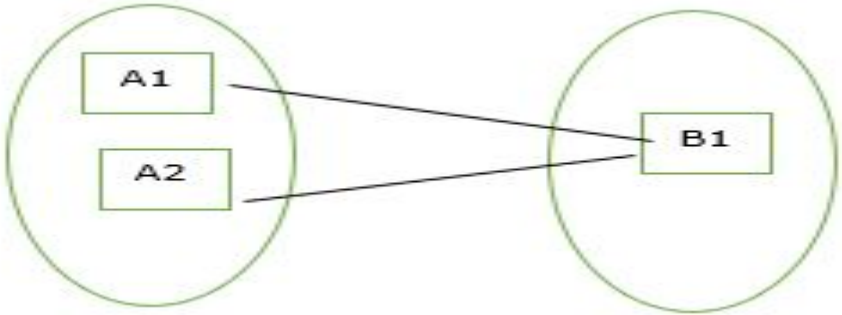
Example

Given below is an example of the one-to-many relationship in the mapping cardinality. Here, one department has many faculties.



Many-to-one relationship

An entity set A is associated with at most one entity in B and an entity set in B can be associated with any number of entities in A with a possibility of zero.



Example

Given below is an example of the many-to-one relationship in the mapping cardinality. Here, many faculties work in one department.

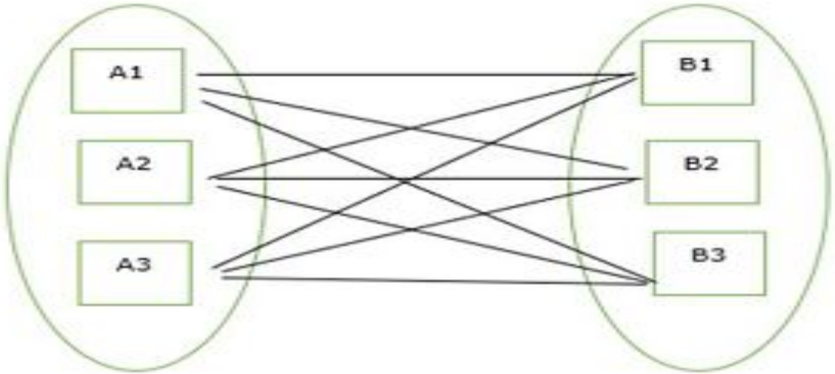


Many-to-many relationship

Many entities of A are associated with many entities of B.

An entity in A is associated with many entities of B and an entity in B is associated with many entities of A.

Many to many=many to one + one to many



Example

Given below is a g cardinality. Here, many employees work on many projects.



Keys in DBMS (Database Management System)

Keys are fundamental concepts in database design that help maintain data integrity and establish relationships between tables. Here are the main types of keys in DBMS:

1. Primary Key

- Uniquely identifies each record in a table
- Cannot contain NULL values
- Only one primary key per table
- Example: `student_id` in a Students table

2. Foreign Key

- A field (or collection of fields) in one table that refers to the primary key in another table
- Maintains referential integrity between tables
- Example: `dept_id` in Employees table that references `dept_id` in Departments table

3. Candidate Key

- A column or set of columns that could qualify as a primary key
- All candidate keys have unique values and no NULLs
- Example: In a Students table, both `student_id` and `email` could be candidate keys

4. Super Key

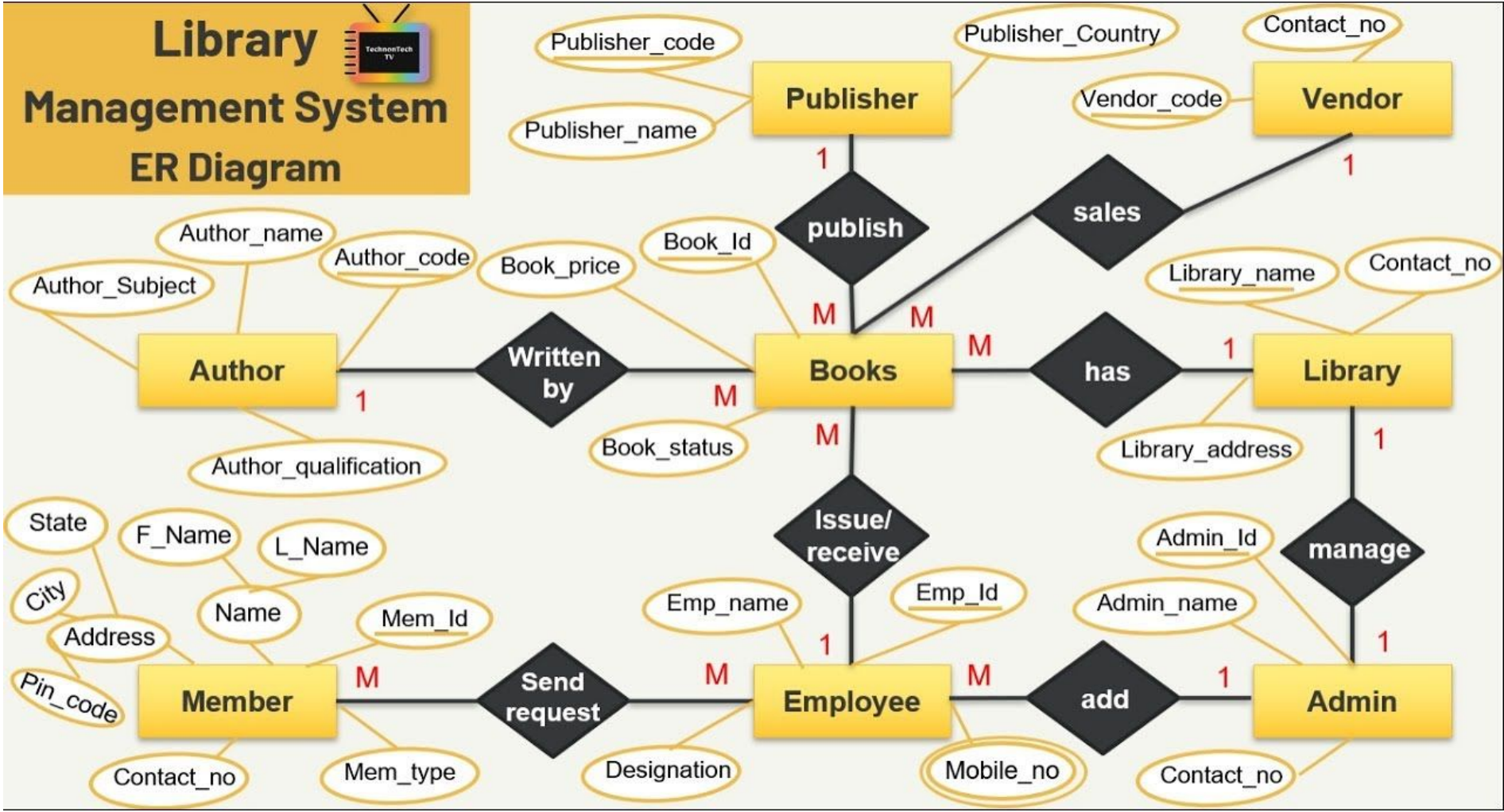
- A set of one or more columns that can uniquely identify a record
- May contain extra attributes that aren't necessary for unique identification
- Example: `{student_id, name}` is a super key when `student_id` alone is sufficient

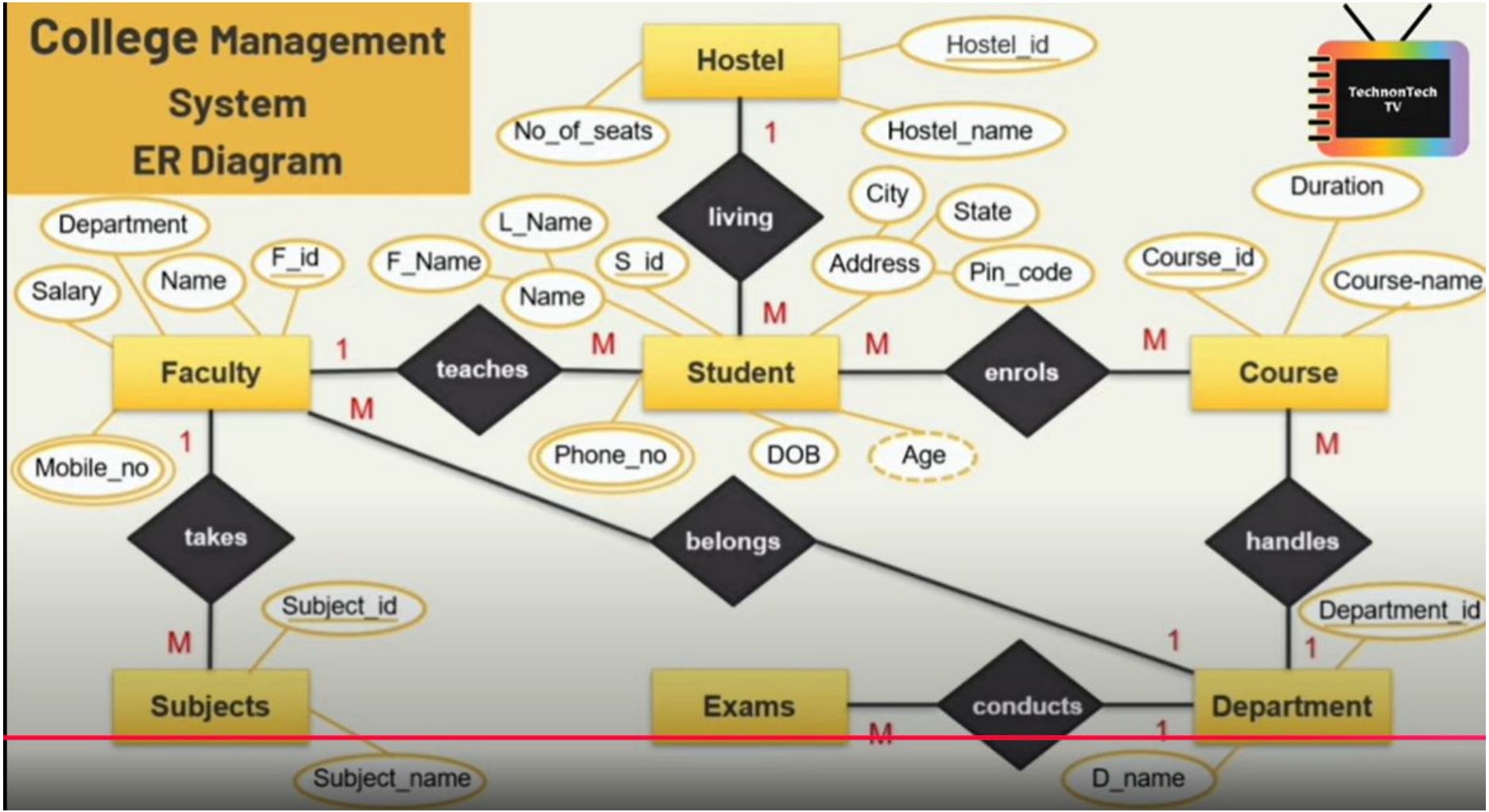
5. Alternate Key

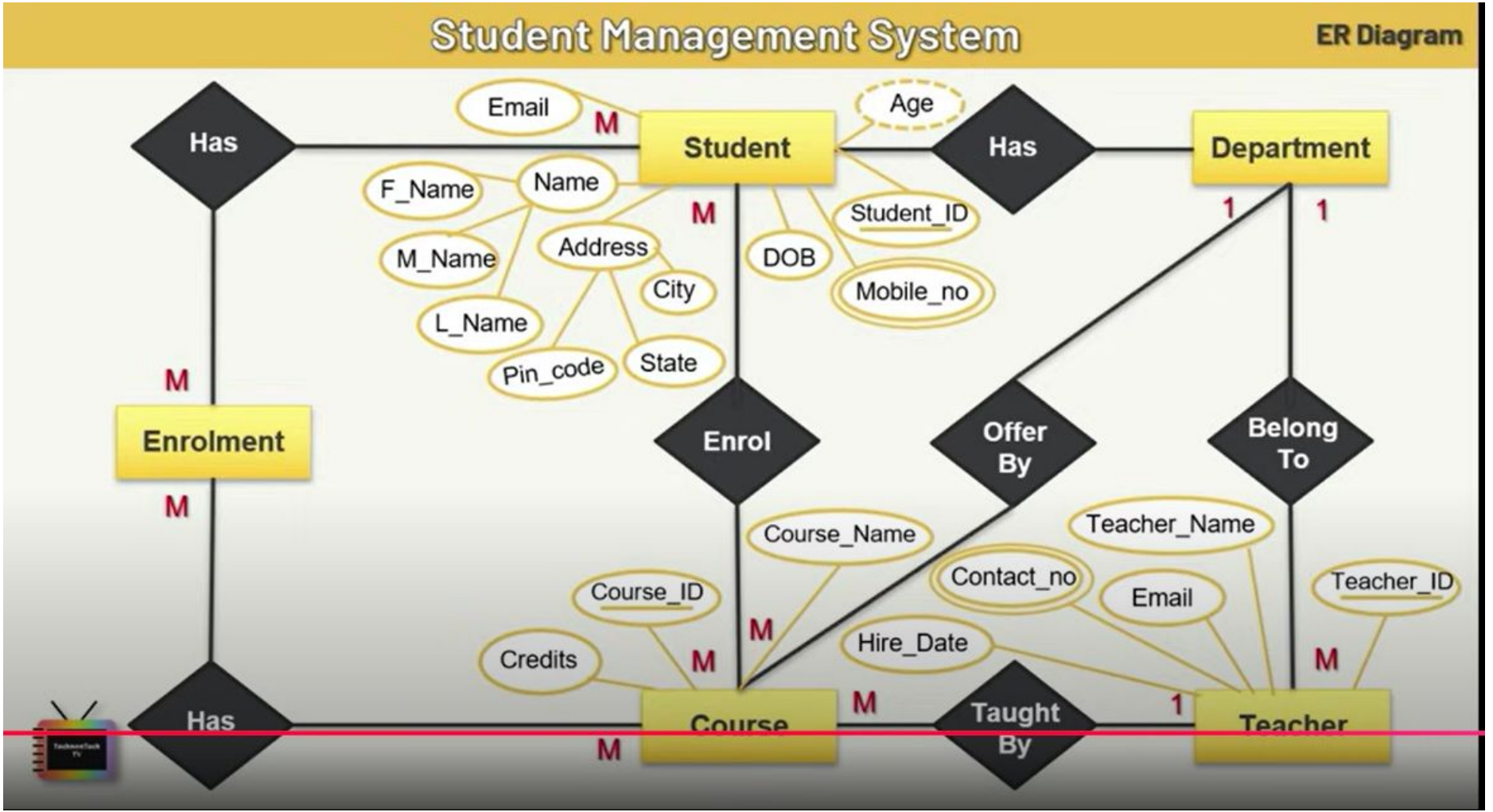
- Candidate keys that are not selected as primary keys
- Example: If `student_id` is primary key, then `email` becomes an alternate key

6. Composite Key

- A primary key that consists of two or more columns
- Used when no single column can uniquely identify a record
- Example: `{order_id, product_id}` in an Order_Details table


Library
Management System
ER Diagram
TechnonTech TV
Publisher_code
Publisher_name
Publisher
Publisher_Country
Contact_no
Vendor_code
Vendor
publish
sales
1
1
Author_name
Author_code
Author_Subject
Author
Author_qualification
Written by
Book_price
Book_Id
Books
Book_status
M
M
M
M
M
1
1
has
Library_name
Contact_no
Library
Library_address
1
State
F_Name
L_Name
City
Name
Address
Pin_code
Mem_Id
Member
Contact_no
Mem_type
Issue/ receive
Send request
Emp_name
Emp_Id
Employee
Designation
Mobile_no
add
Admin_Id
manage
Admin_name
Admin
Contact_no
1
1
1


College Management System ER Diagram
Hostel
Hostel_id
No_of_seats
Hostel_name
1
living
M
City
State
Address
Pin_code
Department
Salary
Name
F_id
L_Name
F_Name
Name
S_id
Faculty
1
teaches
M
Student
M
enrols
M
Course
Duration
Course_id
Course-name
M
1
Mobile_no
takes
M
Subject_id
Subjects
Phone_no
DOB
Age
belongs
handles
1
Department_id
1
Exams
M
conducts
1
Department
Subject_name
D_name
TechnonTech TV


Student Management System
ER Diagram
Has
Email
M
Student
Age
Has
Department
F_Name
Name
M
Student_ID
1
1
M_Name
Address
DOB
City
Mobile_no
L_Name
Pin_code
State
M
Enrolment
Enrol
Offer By
Belong To
M
Course_Name
Teacher_Name
Contact_no
Email
Teacher_ID
Course_ID
M
Hire_Date
Credits
M
M
Has
Course
M
Taught By
1
Teacher
M